

LUKE LOZENSKI

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SUMMARY

Ph.D. Candidate in Systems Science and Mathematics at Washington University in St. Louis. My dissertation work aims at advancing dynamic and multispectral photoacoustic and ultrasound computed tomography through the development of large-scale computational imaging and numerical optimization methods. My research includes the development of scientific machine learning frameworks, such as neural fields and convolutional neural networks, for solving inverse problems.

EDUCATION

Washington University in St. Louis, McKelvey School of Engineering

May 2025

Ph.D. in Science in Systems Sciences and Mathematics

Advisor: Umberto Villa

Secondary Advisory: Joseph O'Sullivan

Washington University in St. Louis, McKelvey School of Engineering

May 2020

Master of Systems Sciences and Mathematics

Washington University in St. Louis, McKelvey School of Engineering

May 2020

Bachelor of Science in Systems Sciences and Engineering

Second Major in Mathematics (Applied Track), School of Arts and Sciences

Minor in Electrical Engineering

TRAINING

Summer Internship at Los Alamos National Laboratory 2024

Summer Internship at Los Alamos National Laboratory 2023

Applied Machine Learning Summer Fellowship from Los Alamos National Laboratory 2022

Independent Study in Computational Chemistry at University of St. Mary, 2016

HONORS AND AWARDS

Imaging Science Pathway Fellow NIH funded training fellowship, 2022-2024

Applied Machine Learning Summer Fellowship from Los Alamos National Laboratory 2022

Distinction in Mathematics Awards from the Washington University in St. Louis Math Department 2020

Dean's List: *Recognition of exceptional undergraduate scholarship* Fall 2016, Spring 2017, Spring 2019, Fall 2019

JOURNAL PUBLICATIONS

1. **L. Lozenski**, R. M. Cam, M. D. Pagel, M. A. Anastasio and U. Villa, *ProxNF: Neural Field Proximal Training for High-Resolution 4D Dynamic Image Reconstruction*, in IEEE Transactions on Computational Imaging, vol. 10, pp. 1368-1383, 2024, doi: 10.1109/TCI.2024.3458397) [[journal](#)]
2. **L. Lozenski**, H. Wang, F. Li, M. Anastasio, B. Wohlberg, Y. Lin, U. Villa, *Learned Full Waveform Inversion Incorporating Task Information for Ultrasound Computed Tomography*, IEEE Transactions on Computational Imaging (2024) [[journal](#)]
3. B. Liang, J. Tan, **L. Lozenski**, D. Hormuth, T. Yankeelov, U. Villa, D. Faghihi, *Bayesian Inference of Tissue Heterogeneity for Prediction of Glioma Growth*, IEEE Transactions on Medical Imaging (2023) [[journal](#)]
4. **L. Lozenski**, M. Anastasio, U. Villa, *A Memory-Efficient Self-Supervised Dynamic Image Reconstruction Method using Neural Fields*, IEEE Transactions on Computational Imaging (2022) [[journal](#)]
5. M. Watanabe, and **L. Lozenski**, *Thermodynamics study of absorption of aromatic organic compounds to carbon nanotubes*, Abstracts of Papers of the American Chemical Society. Vol. 251. 1155

MANUSCRIPTS IN PREPARATION

6. **L. Lozenski**, R. M. Cam, M. D. Pagel, M. A. Anastasio and U. Villa, *In-Vivo Dynamic Contrast Enhanced Photoacoustic Computed Tomography utilizing Neural Field Representations*
7. **L. Lozenski**, M. McCann, B. Wohlberg, *An Adaptive multiparameter Penalty Method for ADMM Applied to Multiconstraint and Multiblock Optimization Problems*
8. **L. Lozenski**, H. Wang, F. Li, M. Anastasio, B. Wohlberg, U. Villa, Y. Lin, *Learned Correction Methods for Simplified Acoustic Forward Models in Ultrasound Computed Tomography*

CONFERENCE PROCEEDINGS

9. **L. Lozenski**, H. Wang, B. Wohlberg, U. Villa, Y. Lin, *Learned measurement correction for simplified acoustic forward models in ultrasound computed tomography*, Medical Imaging 2024: Ultrasonic Imaging and Tomography. Vol. 12932. SPIE, 2024.
10. **L. Lozenski**, H. Wang, B. Wohlberg, U. Villa, Y. Lin, *Data-Driven Methods for Ultrasound Computed Tomography*, Medical Imaging 2023: Physics of Medical Imaging. Vol 12463. SPIE, 2023.
11. **L. Lozenski**, M. Anastasio, U. Villa, *Neural Fields for Dynamic Imaging*, Medical Imaging 2022: Physics of Medical Imaging. Vol. 12031. SPIE, 2022.

TECHNICAL REPORTS

12. **L. Lozenski**, R.M. Cam, M. Anastasio, U. Villa, *An Efficient Implementation of the Spherical Radon Transform with Cylindrical Apertures*, ArXiv Preprint arXiv:2402.15641 (2024) [[preprint](#)]
13. B. Liang, **L. Lozenski**, U. Villa, D.I Faghihi, *PDE-constrained optimization formulation for tumor growth model calibration*, arXiv preprint arXiv:2302.06445 (2023) [[preprint](#)]
14. **L. Lozenski**, U. Villa, *Consensus ADMM for inverse problems governed by multiple PDE Models*, ArXiv Preprint arXiv:2104.13899 (2021) [[preprint](#)]

ORAL PRESENTATIONS

1. *Learned measurement correction for simplified acoustic forward models in ultrasound computed tomography*, SPIE Medical Imaging 2024, San Diego, CA, US February 19, 2024,
2. *Accurate Efficient Multispectral Spatiotemporal Image Reconstruction Using Neural Field Representations*, Electronic Imaging 2024, San Francisco CA, US, January 25, 2024
3. *Proximal Splitting Methods for Neural Field Based Dynamic-Image Reconstruction*, SIAM Optimization 2023, Seattle, WA, US, June 2, 2023
4. *Proximal Splitting Methods for Neural Field Based Dynamic-Image Reconstruction*, WashU Imaging Science Retreat 2023, St. Louis MO,, US, March 30, 2023
5. *Data-driven Methods for Ultrasound Computed Tomography*, SPIE Medical Imaging 2023, San Diego, CA, US February 21, 2023,
6. *Neural Fields for Dynamic Imaging*, SPIE Medical Imaging 2022, San Diego, CA, US February 23, 2022
7. *Consensus ADMM for Inverse Problems Governed by Multiple PDE Models*, FEniCS 2021 Conference Proceedings (pg. 749). Virtual. March 22- 26, 2021

POSTER PRESENTATIONS

8. **Lozenski, L.**, Villa, U., *Neural Fields for Dynamic Imaging*, WashU Imaging Sciences Pathway Retreat 2022
9. **Lozenski, L.**, Villa, U., *Implicit Neural Representations for Dynamic Photoacoustic Tomography Imaging*, WashU ESE Research Showcase 2021
10. Datta, A., **Lozenski, L.**, Xiao J., Villa, U., *Learning from data through the lens of mathematical models*, WashU ESE Research Showcase 2020

RESEARCH EXPERIENCE

In-Vivo DCE-PACT utilizing Neural Fields

June 2022-

- Developed a 5D neural field image reconstruction method for dynamic, multispectral photoacoustic tomography imaging
- Systematically assessed by use of virtual imaging studies featuring high-resolution anatomically realistic numerical phantoms and high fidelity computational models of the data acquisition (diffusion approximation for photon transport and wave equation for acoustic propagation)
- Applied to in-vivo small-animal imaging to estimate tumor perfusion
- Combines state of the art machine learning approaches with traditional model-based reconstruction methods

Proximal Splitting Methods for Neural Field Based Dynamic-Image Reconstruction March 2021-Jun2-2024

- Development of scalable, memory-efficient dynamic image reconstruction algorithms using neural fields to represent 5D images
- Application to photoacoustic dynamic imaging (circular and spherical Radon transform)
- Implementation of algorithm in PyTorch and comparison with compressed sensing (low-rank based dynamic image reconstruction algorithms)

Adaptive Penalty Parameter Methods for Multiconstraint ADMM

June 2024-

- Work done during summer internship at Los Alamos National Laboratory under supervision of Brendt Wohlberg and Michael McCann
- Development and rigorous analysis of adaptive selection of multiple penalty parameter of the alternating direction method of multipliers (ADMM)
- Automatically adjusts for scale differences between constraints and multiblock subproblems
- Designed to solve a wide variety of constrained optimization problems with particular relevance to image reconstruction problems

Learned Measurement Correction for Simplified Acoustic Forward Models in USCT

June 2023-

- Work done during summer internship at Los Alamos National Laboratory under supervision of Youzuo Lin
- Development of learning approaches to compensate for modeling errors in the solution of inverse problems: An application to USCT image reconstruction under the Born approximation
- The proposed methods were systematically assessed by use of anatomically and physiologically realistic numerical phantoms and objective measurement of image quality from statistical signal detection theory

Learned Full Waveform Inversion Incorporating Task Information for USCT

June 2022-August 2023

- Applied Machine Learning Fellowship at Los Alamos National Laboratory under supervision of Youzuo Lin
- Development of convolutional neural network for data driven solutions to USCT inverse problems
- Learned reconstruction methods tailored for specific engineering tasks with task-informed training
- Machine learning framework using PyTorch on GPU and HPC clusters

Inversion Algorithms for Quantitative Photoacoustic Tomography

June 2019-

- Development of techniques for image reconstruction of biological tissue optical properties (absorption and scattering coefficient)
- Python implementation utilizing Fenics and hiPPylib for solving inverse problems
- Collaboration with Dr. Mark Anastasio at University of Illinois at Urbana-Champaign

Consensus ADMM for Inverse Problems Governed by Multiple PDE Forward Models May 2020-August 2021

- Solution methods for inverse problems with multiple PDE forward models and non-smooth regularization
- Implemented the alternating direction method of multipliers (ADMM) with demonstrated scalability with respect to problem size and number of PDE forward models

Bayesian Inference of Tissue Heterogeneity for Glioma Growth

May 2020-April 2023

- Investigation on modeling tumor growth under the effects of radiotherapy as a time-dependent PDE
- Collaboration with Dr. Danial Faghihi at the University at Buffalo.

TEACHING EXPERIENCE

Washington University

<i>Assistant Instructor – E35 ESE Optimization (Dr. Tunay)</i>	August 2024-December 2024
<i>Assistant Instructor – E35 ESE Large-Scale Optimization (Dr. Kamilov)</i>	August 2024-December 2024
<i>Assistant Instructor – E35 ESE 5931 Mathematics of Imaging Science (Dr. Jha)</i>	August 2022-December 2022
<i>Assistant Instructor – E35 ESE 524 Detection and Estimation Theory (Dr. Nehorai)</i>	January 2022-May 2022
<i>Assistant Instructor – E35 ESE 520 Probability and Stochastic Processes (Dr. Kurenok)</i>	January 2021-May 2021
<i>Grader – E35 ESE 105 Introduction to Systems Engineering (Dr.'s Ching and Lew)</i>	August 2019-December 2019
<i>Grader – L24 Math 4111 Introduction to Analysis (Dr. Shareshian)</i>	August 2019-December 2019
<i>Grader – L24 Math 5045 Geometry/Topology I: Algebraic Topology (Dr. Roberts)</i>	August 2019-December 2019
<i>Grader – L24 Math 430 Modern Algebra (Dr. Bongers)</i>	January 2020-May 2020

GRADUATE LEVEL RELEVANT COURSEWORK

Optimization

- Large-Scale Optimization for Data Science E81 CSE 534A
- Optimization and Optimal Control E35 ESE 544
- Convex Optimization E35 ESE 519

Imaging Science

- Biological Imaging Technologies E62 BME 589
- Mathematics of Imaging Science E35 ESE 5931
- Computational Methods for Imaging Science E35 ESE 5932
- Theoretical Imaging Science E35 ESE 5933
- Practicum in Imaging Science E35 ESE 5934

Probability and Statistics

- Detection and Estimation Theory E35 ESE 524
- Probability and Stochastic Processes E35 ESE 520
- Advanced Systems Science for Learning and Control of Stochastic Dynamic Systems ESE 5581

Mathematics and Systems Analysis

- Data-Integrated Frameworks for Systems Analysis and Control Design E35 ESE 559
- Linear/Nonlinear Dynamic Systems E35 ESE 551-552
- Advanced Systems Science for Learning and Control of Dynamic Systems E35 ESE 5581
- Harmonic Analysis L24 Math 519
- Geometry/Topology L24 Math 5045- 5047
- Measure Theory and Functional Analysis L24 Math 5051-5052
- Sobolev Spaces and PDEs L24 Math 523

COMPUTER SKILLS

- Expert in Python and skilled in use of a wide variety of packages
 - Expert knowledge of PyTorch, an open source machine learning library
 - Experience with Fenics, a parallel finite element program for solving PDEs
 - Experience with Devito, an optimized finite difference program for solving PDEs
 - Skilled with hIPPYlib, a computational framework for solving inverse problems.
- Expert with MATLAB for numerical methods and data science
- Linux shell, computer clusters, and MPI parallel jobs
- Latex

SERVICE

Peer review: contributed to peer review of scientific manuscripts for *Photacoustics* and *SoftwareX*

Graduate Student Advisory Board: Served on advisory board for graduate students in ESE department

Student Welcome Days: Lead and assisted events for admitted Ph.D. students and introduction events

PROFFESIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE) (*since 2021*)

Society for Industrial and Applied Mathematics (SIAM) (*since 2021*)

International Society for Optics and Photonics (SPIE) (*since 2021*)

REFERENCES

Dr. Umberto Villa, Research Associate Professor, Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin uvilla@oden.utexas.edu *Ph.D. advisor*

Dr. Brendt Wohlberg, Scientist, Computer, Computational, and Statistical Sciences Division, Los Alamos National Laboratory brendt@ieee.org *Internship Advisor*

Dr. Ulugbek Kamilov, Associate Professor, Electrical & Systems Engineering , Washington University in St. Louis kamilov@wustl.edu *Instructor and PhD committee member*

Dr. Youzuo Lin, Associate Professor, School of Data Science, University of North Carolina at Chapel Hill yzlin@unc.edu *Internship Advisor*